The Conditional Activation Model: Uncovering Policy Demand Responses to AI-Driven Automation

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Outline

1. Puzzle & Theory
2. Research Design
3. Identification & Model
4. Results
5. Discussion

Puzzle & Theory

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⇒ **When—and for whom—does exposure to automation risk translate into policy demand?**

Mixed Findings in the Literature

Study	Trust Modeled?	Prior Beliefs Elicited?	Engagement Required?	Policy Framing	Finding
Zhang (2022)	No	No	No (vignette)	Mixed	Null
Magistro et al. (2024)	No	No	No (conjoint)	Contextualized	Null
Ladreit (2022)	No	No	No (vignette)	Contextualized	Null
Arntz et al. (2024)	No	Yes (optimism)	No	Traditional	**Heterogeneous**
Heinrich & Witko (2024)	Proxy (knowledge)	No	No	Mixed	**Conditional+**
Haslberger (2024)	Reduced (behavioral)	No	Yes (AI use)	Traditional	**Positive**
This Study	**Yes (latent)**	**Yes (Gap)**	**Yes (3-step)**	**Contextualized**	**CATE**

Studies without trust modeling or belief elicitation $\rightarrow$ null effects.

Design features accumulate $\rightarrow$ conditional effects emerge.

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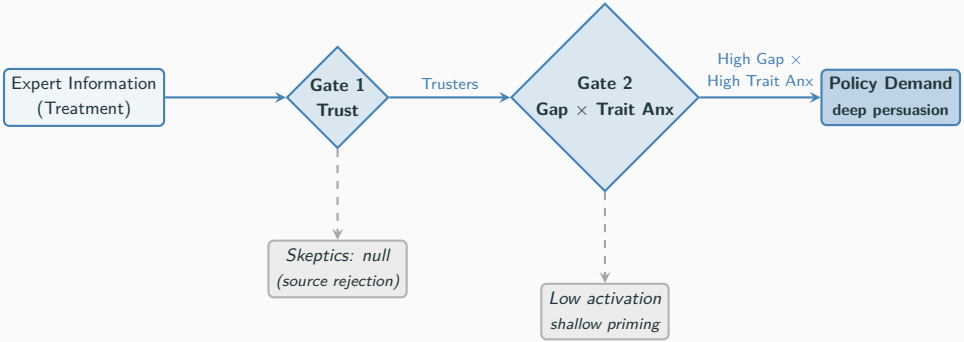
- High-disposition individuals override partisan heuristics and engage effortfully with threatening information *(Albertson & Gadarian, 2015)*.
- **Belief Gap** = expert estimate − prior belief: the "activation potential" for updating.

The Conditional Activation Model



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H5. Conditional Activation. Largest effects among Trusters with high anxiety *and* positive Gaps—both gates satisfied.

The Right Estimand

Why the ATE understates the story:

$$\text{ATE} = \underbrace{\pi}_{\text{Truster share}} \cdot \tau_{\mathrm{T}} + \underbrace{(1-\pi)}_{\text{Skeptic share}} \cdot \underbrace{\tau_{\mathrm{S}}}_{\text{flat}}$$

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If τ_{S} is positive but **unmoderated** (no response to Gap or Anxiety), the ATE **dilutes** the conditional signal. The theoretically relevant quantity is the **CATE**.

Research Design

Sample and Randomization Balance

Sample

- $N = 901$ U.S. workers via CloudResearch Connect (Treatment: 475 / Control: 426)
- Targeting (enforced): ages 18–65; full-time, part-time, business owner, or unemployed; English-speaking

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Manipulation checks

- 93.4% comprehension; perceived gap tracks objective gap ($\rho = 0.73$); belief updating tracks gap ($\rho = 0.66$)

Treatment: Three-Step Belief Correction

Step 1: Neutral context

Automation and Your Job

Artificial intelligence (AI) is getting better at doing work people used to do. Computers already answer customer questions, write basic code, and sort medical images. *As AI improves, some jobs may be partly-or fully-done by machines, while new kinds of jobs also appear.*

How do experts judge a job's automation risk?

Experts look at the small tasks that make up a job. When those tasks are routine, clearly defined, and follow repeated patterns-and there are lots of examples for computers to learn from-AI systems can pick them up more easily.
The shorter and more repetitive a task is, the easier it is to automate.

EXAMPLE JOB	TYPICAL TASK	AI RISK (TODAY)
IT Help Desk Technician	10-minute help IT tickets that follow a script	*Higher Risk* – chatbots can handle most tickets
IT Systems Designer	Plan how the company's tech infrastructure will work over the next year	*Lower Risk* – needs long-term planning and team buy-in

Other factors also influence how likely a job is to be replaced by AI automation:

Jobs are at greater risk when they:

- Can be done entirely online, or
- Have lots of recorded data for AI to learn from.

Jobs are less at risk when they:

- Require in-person skills or dexterity, or
- Rely on human trust or empathy.

Step 2: Elicit prior belief

Imagine every worker in the country standing on one long line, sorted by how likely their job is to be automated.
The very front of the line (position 0) belongs to people whose jobs are safer than almost everyone else's, while the very back (position 100) belongs to those whose jobs are at risk sooner than almost everyone else's.
Someone standing in the middle of the line (position 50) would have half the number of workers ahead of them and half behind.

- **Safest** jobs are those whose chances of being automated are below most other people.
- **Riskier** jobs are those with an automation potential above most other people.

Taking a moment to reflect on your own job, where do you feel you would place yourself on that line right now?

0=Safest than everyone else's. Riskier than everyone else's=100.
0 10 20 30 40 50 60 70 80 90 100

Click and drag on the slider to select your occupation's relative risk of automation.

Step 3: Reveal expert & gap

Comparing Your Job Automation Risk: Your View vs. Expert Insight

Earlier, we asked you to imagine all workers standing in a single line, ranked by the risk of their jobs being automated — from the lowest relative risk (position 0) to the highest relative risk (position 100).

You placed yourself at: **Position 37**

Researchers place your job at around: **Position 82**

This means that, according to experts, your job has a **higher** risk of automation than **82** out of 100 people — but a **lower** risk than **18** others.

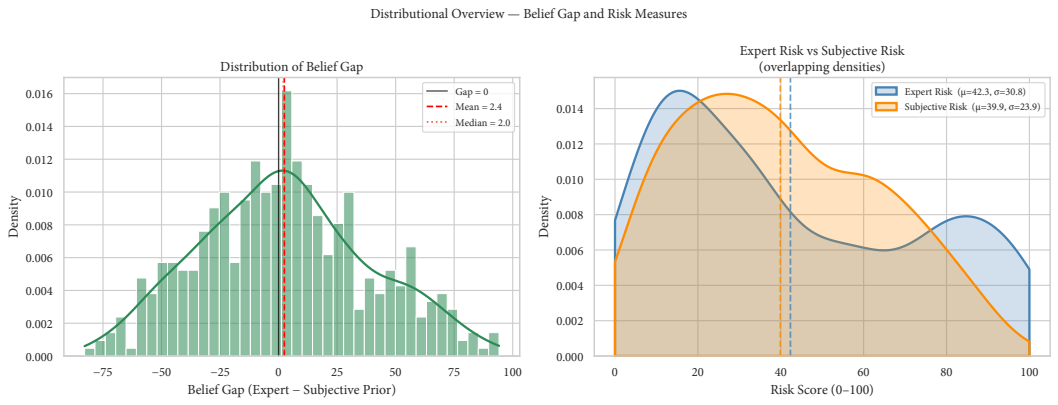
Regarding how likely is your job to be automated in the future relative to others, how does the experts automation risk assessment compare to your initial view?

- ○ The experts place my job at a **significantly higher** relative risk
- ○ The experts place my job at a slightly higher relative risk
- ○ The experts place my job at a fairly similar relative risk
- ○ The experts place my job at a slightly lower relative risk
- ○ The experts place my job at a **significantly lower** relative risk

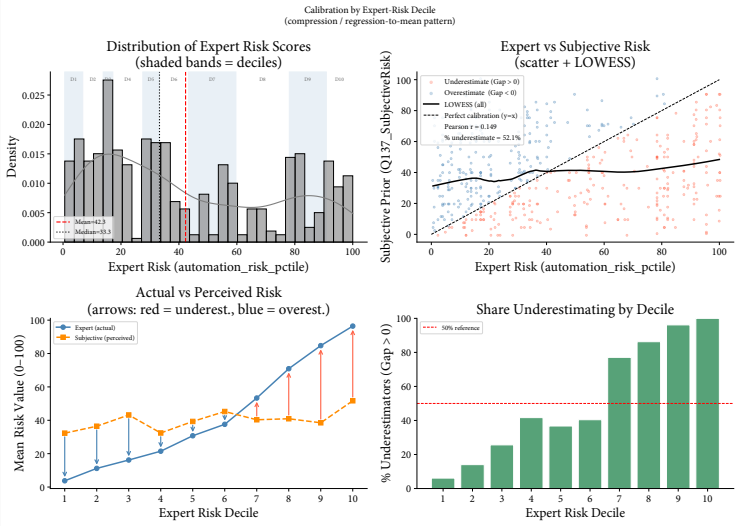
Control group: skips all steps; proceeds directly to outcome battery.

Belief Gap = Expert estimate − Prior belief (computed, not shown to participants).

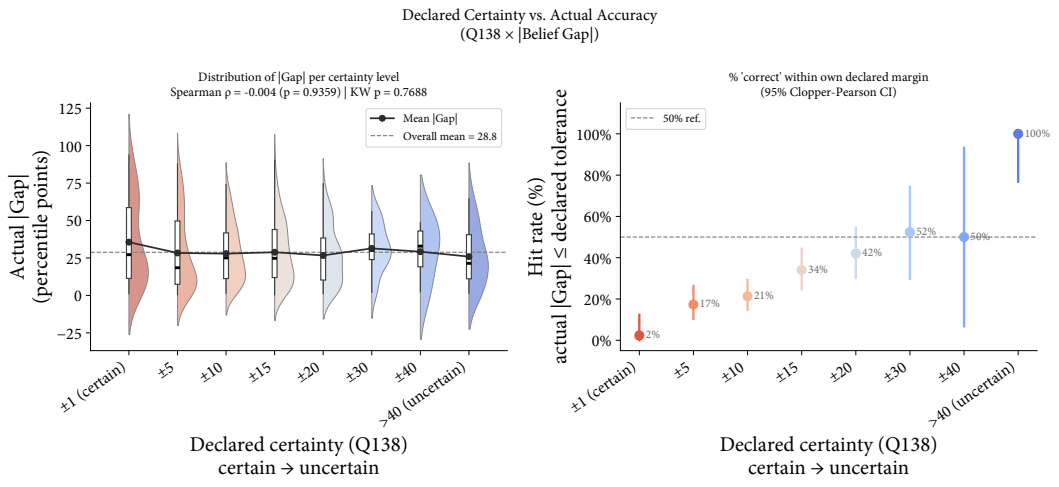
The Belief Gap: Construction and Distribution



Belief Gap: Systematic Miscalibration



Belief Gap: Workers Don't Know How Wrong They Are



Manipulation Checks

Perceived Gap Direction (Q141)

Category	n	Mean Gap
Sig. higher risk	143	35.4
Slightly higher risk	98	6.2
Fairly similar risk	71	-2.0
Slightly lower risk	73	-14.4
Sig. lower risk	97	-33.4
Spearman $\rho = -0.733 ***$		

Belief Updating (Q143)

Category	n	Mean Gap
Much lower than prior	55	-35.6
A little lower than prior	78	-23.5
About the same as prior	172	3.4
A little higher than prior	93	15.6
Much higher than prior	84	35.6
Spearman $\rho = 0.656 ***$		

Comprehension check accuracy: 93.4% (treated + control).

Measurement: Scales and Psychometric Properties

Outcome

- **Policy Demand** (Q96): 5 items, 7-pt scale ($\alpha = .82$, $\omega = .82$)

  Unemployment benefits, AI regulation, retraining, labor protections, AI job limits

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- **Policy Demand** (Q96): 5 items, 7-pt scale ($\alpha = .82$, $\omega = .82$)
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Intermediate outcomes

- **Automation Anxiety** (Q94): 4 items, 7-pt scale ($\alpha = .93$, $\omega = .92$)
- **Perceived AI Impact** (Q95): 7 items, 5-pt scale; 2-factor (threat/opportunity)

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Moderators

- **Gate 1 — Trust** (Q142): single item, 5-pt scale (post-treatment, treated only)
- **Gate 2 — Trait Anxiety** (Q23): 4 items, 7-pt scale ($\alpha = .91$, $\omega = .91$)
- **Belief Gap**: Expert percentile − Subjective prior (continuous, 0–100)

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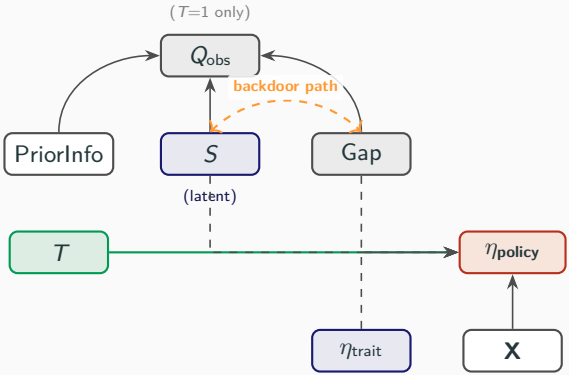
Joint 6-factor CFA (23 items): CFI = .958, RMSEA = .054, SRMR = .048

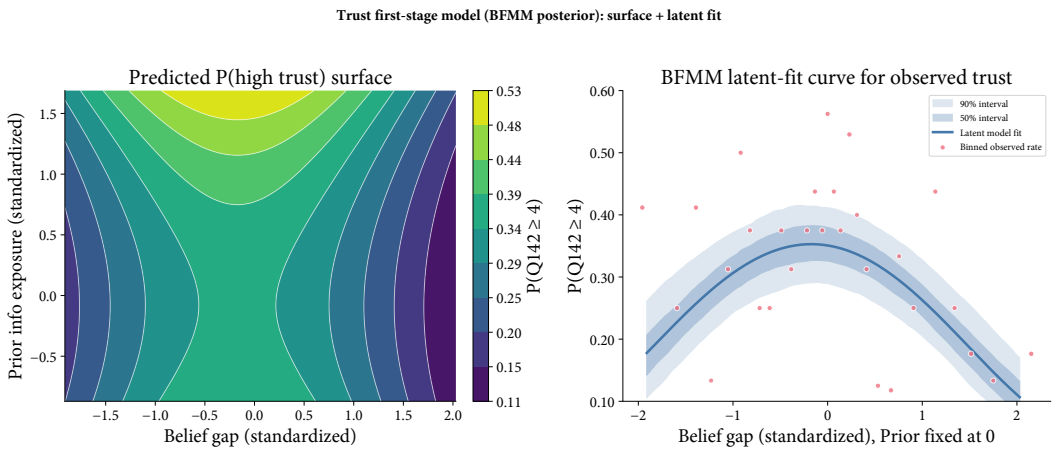
Identification & Model

The Identification Problem: Trust as a Collider



Q_{obs} is a collider: conditioning on observed trust opens a spurious path between Gap and S. $\Rightarrow$ Cannot simply split on observed trust.

Trust Covariates: Epistemic Validation



Bayesian Finite Mixture Model

1. **Latent trust classification** (ordered logit, 2 components)
   - $Q_{\text{obs}} \mid S, \text{Gap}, \text{PriorInfo} \sim \text{OrderedLogit}(\delta S + \gamma' \mathbf{z}, \kappa)$
   - Covariates $\mathbf{z}$: Gap, Gap^2, PriorInfo, PriorInfo^2 (blocks collider)
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3. **Class-specific structural equations**
   - $\tau_{\text{T}} = \beta_0^T + \beta_1^T \cdot \text{Gap} + \beta_2^T \cdot \eta_{\text{trait}} + \beta_3^T \cdot (\text{Gap} \times \eta_{\text{trait}})$
   - $\tau_{\text{S}} = \beta_0^S + \ldots$ (same form, different parameters)
   - Constraint: $\beta_\Delta \geq 0$ (Trusters $\geq$ Skeptics monotonicity)

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Estimation: Stan/HMC-NUTS, 4 chains $\times$ 2,000 draws. All $\hat{R} < 1.01$.

Structural Model (Gate 2)

Latent policy demand decomposes into structural mean + residual:

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The CATE for each class:

$$\tau_S(\text{Gap},\ \eta_{\text{trait}}) = \beta_{S,1} + \beta_{S,2} \cdot \text{Gap} + \beta_{S,3} \cdot \eta_{\text{trait}} + \beta_{S,4} \cdot \underbrace{\text{Gap} \times \eta_{\text{trait}}}_{\text{key interaction (H5)}}$$

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Results

Unconditional Average Treatment Effect

Estimator	**ATE (d)**	**SE/SD**	**95% CI**
OLS (no controls)	0.27	0.07	[0.14, 0.40]
OLS (with controls)	0.24	0.06	[0.13, 0.36]
SEM (no controls)	0.26	0.07	[0.13, 0.39]
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BFMM (π-weighted)	0.27	0.06	[0.14, 0.39]

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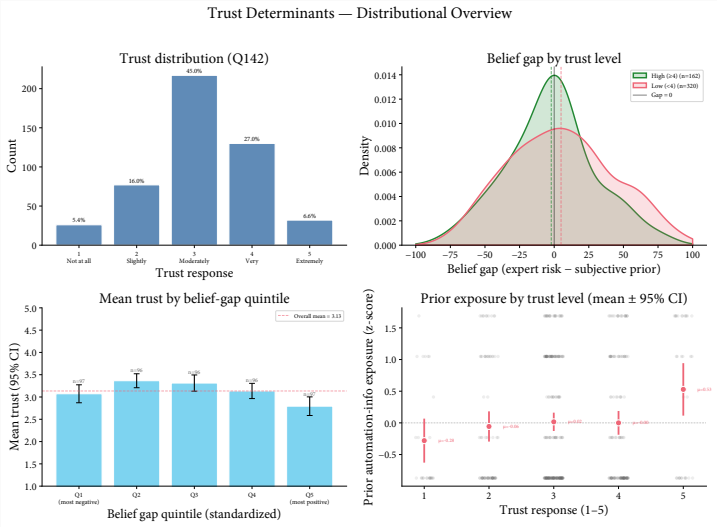
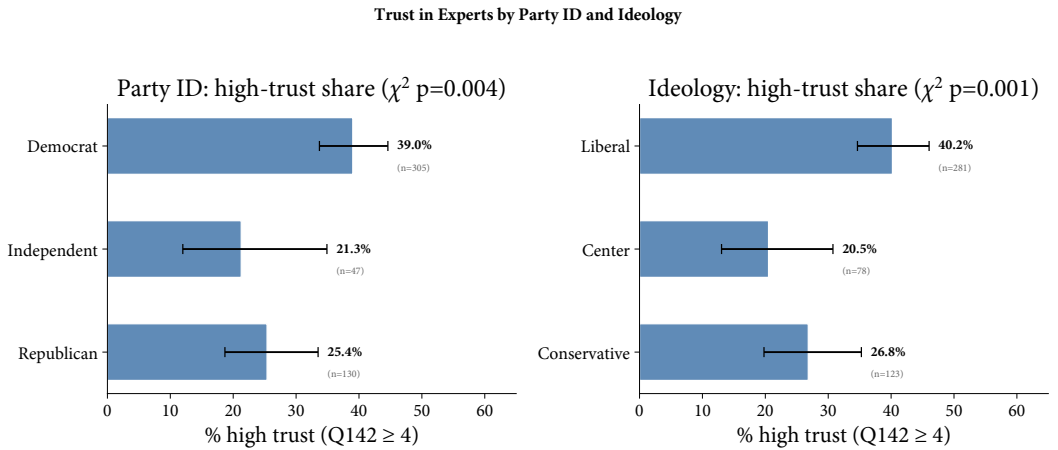
All estimators converge at $d \approx 0.24$–0.27, indicating a positive but modest average treatment effect.

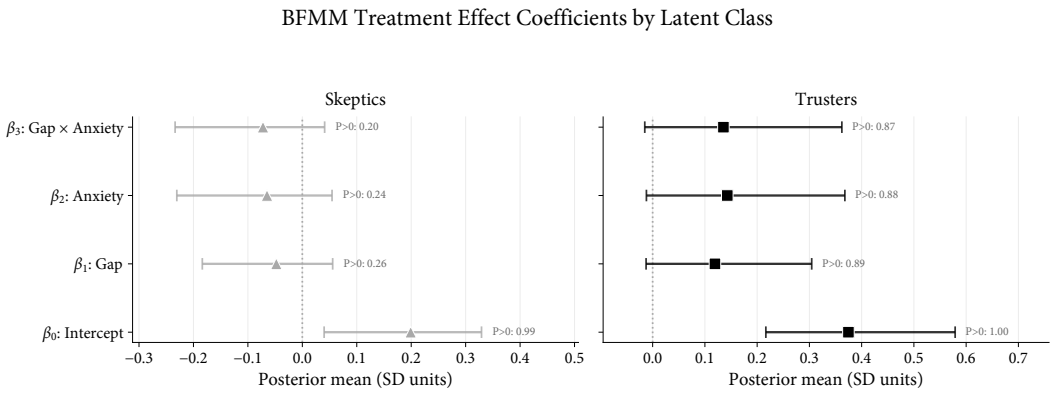
First Stage: Latent Trust Classes



Trust and Partisan Identity

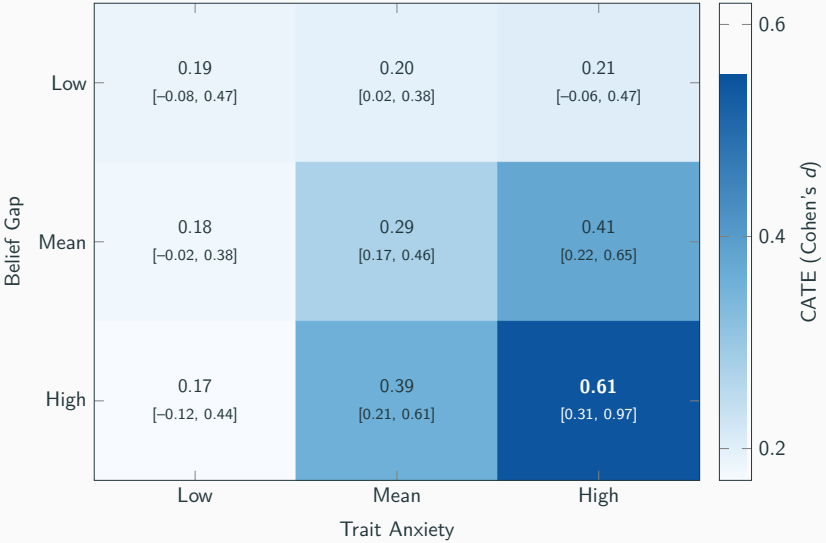


BFMM Coefficients: Skeptics vs Trusters

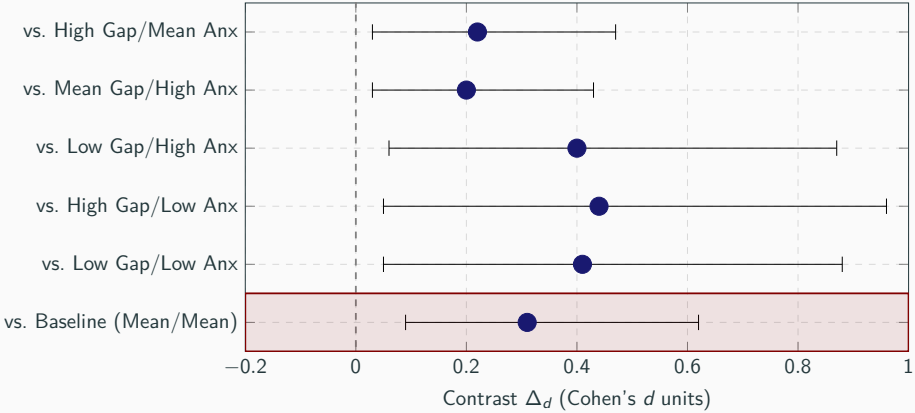


Note: Dots = posterior mean. Bars = 90% credible interval. P>0 = posterior probability coefficient is positive.

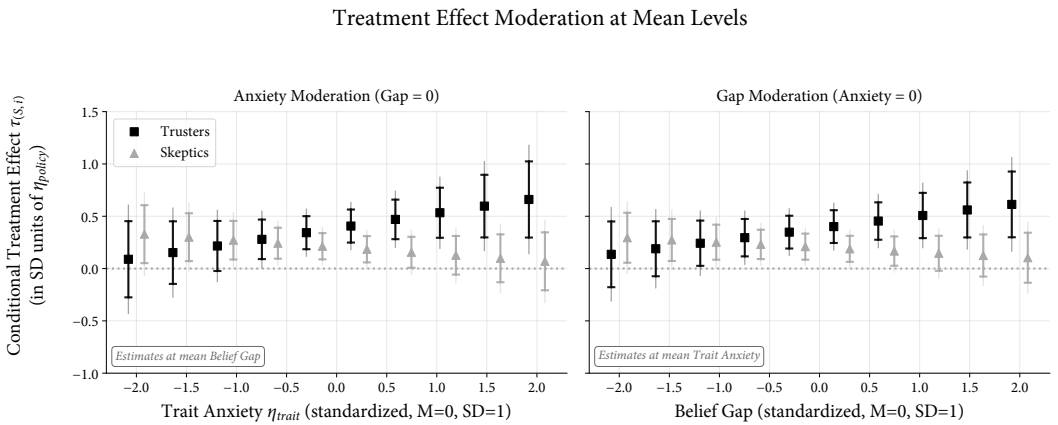
CATE Surface: Treatment Effects Among Trusters



CATE Contrasts: Maximum Activation Dominates

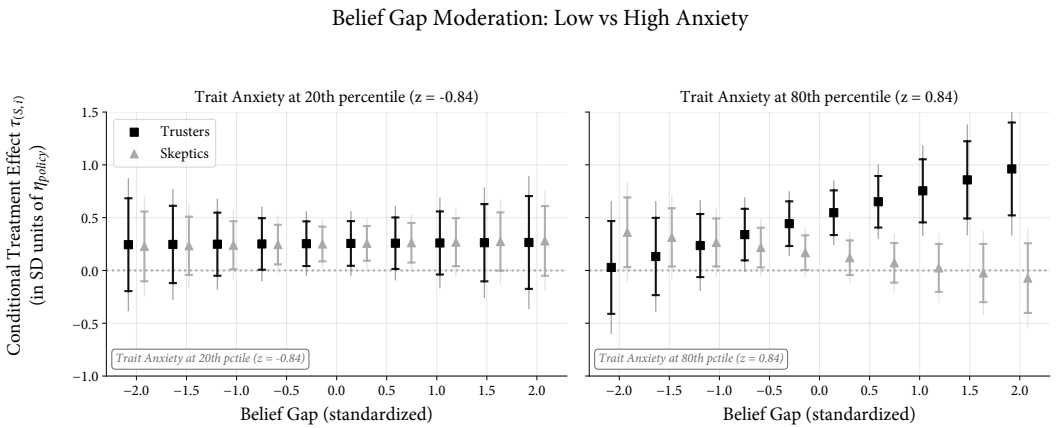


Partial Effects: Gate 1 Is a Strict Filter



Note: Points show posterior mean. Thick bars with caps = 83% CI. Thin lines = 95% CI. Dotted line = null effect.

Belief Gap Moderation: Low vs High Anxiety



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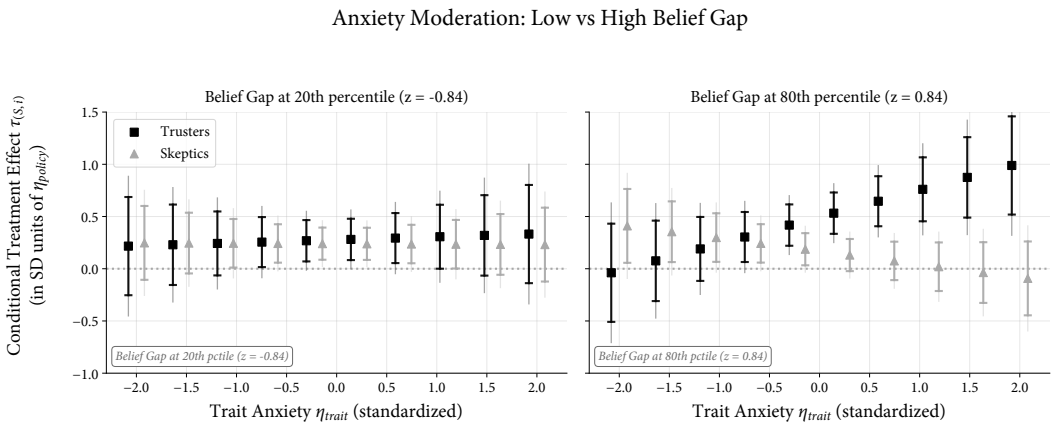
Anxiety Moderation: Low vs High Belief Gap



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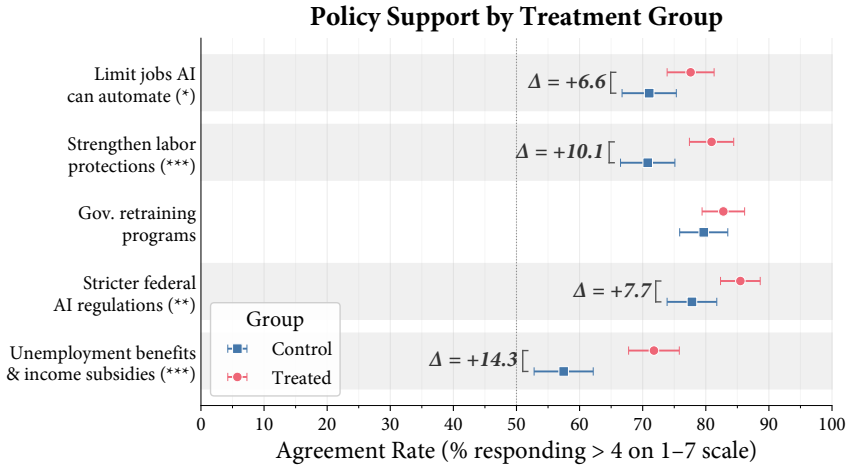
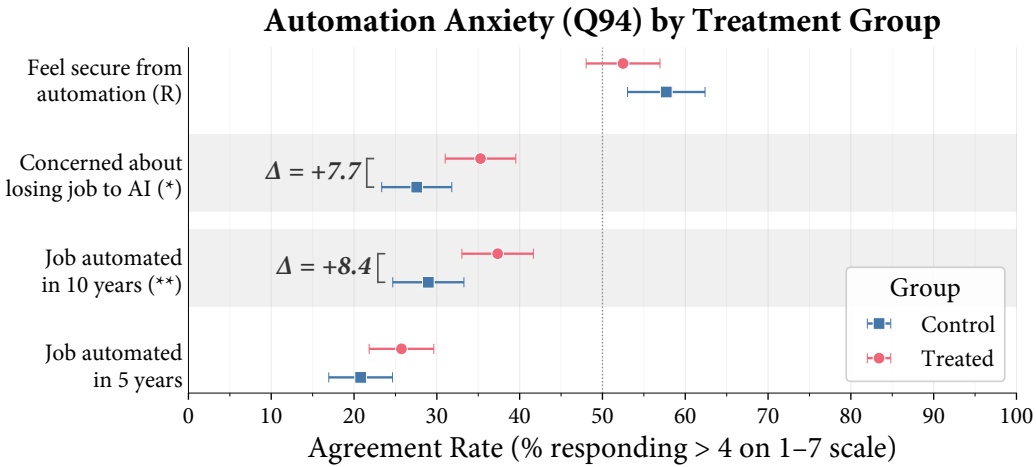
Item-Level Effects: Policy Support (Q96)



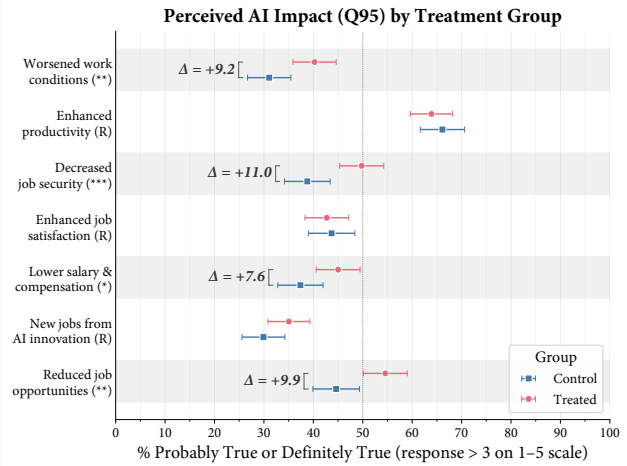
Note: * p < .05, ** p < .01, *** p < .001 (two-proportion z-test). Shaded rows = p < .05.
Δ = Treated − Control (pp). Error bars = 95% Wald CI.

Item-Level Effects: Automation Anxiety (Q94)



Note: * p < .05, ** p < .01, *** p < .001 (two-proportion z-test). Shaded rows = p < .05.
Δ = Treated − Control (pp). Error bars = 95% Wald CI. (R) = reverse-worded item.

Item-Level Effects: Perceived AI Impact (Q95)



Note: * p < .05, ** p < .01, *** p < .001 (two-proportion z-test). Shaded rows = p < .05.
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Discussion

Three Pathways, Not Two

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   τ_S positive at intercept ($d \approx 0.20$) but *flat*: no moderation by Gap or Anxiety.
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The "null vs. works" framing in the literature is a false dichotomy.
All three pathways operate simultaneously within the same experiment.

Limitations

- Single pre-registered experiment — no replication sample
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- Untestable independence assumption: $S \perp \text{Gap}$ (dispositional trust does not cause prior beliefs)
- Policy battery skews toward protective/supportive items — may understate restrictive demand

Takeaway

1. **The ATE masks conditional persuasion.**
   Unconditional $d \approx 0.27$ averages over an unmoderated subgroup (Skeptics, $d \approx 0.20$ flat) and a strongly moderated subgroup (Trusters, d up to 0.61). The estimand matters.

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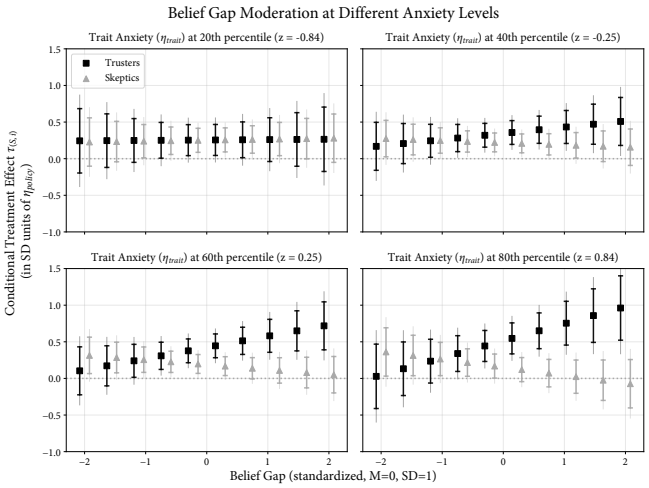
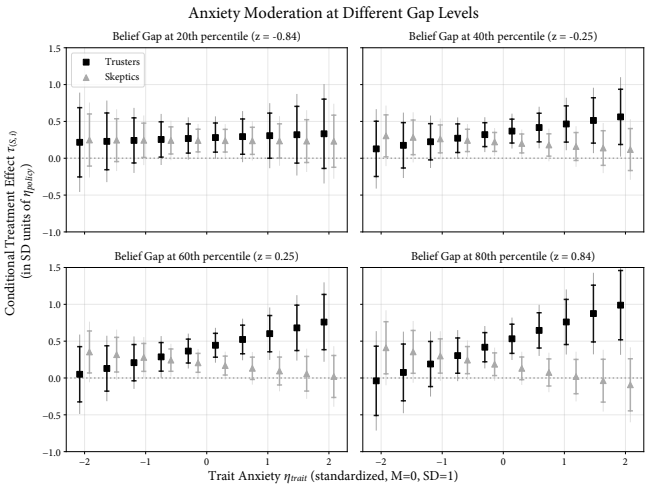
3. **The two-gate architecture may generalize.**
   Source credibility $\times$ information shock $\times$ dispositional anxiety is not automation-specific — same logic applies to climate, vaccines, trade. Open empirical question.

Appendix

Gap Slope Steepens With Anxiety



Anxiety Slope Steepens With Belief Gap



Note: Points show posterior mean. Thick bars with caps = 83% CI. Thin lines = 95% CI. Dotted line = null effect.

Instrument: Neutral Context Vignette (Step 1)

Automation and Your Job

Artificial intelligence (AI) is getting better at doing work people used to do. Computers already answer customer questions, write basic code, and sort medical images. *As AI improves, some jobs may be partly-or fully-done by machines, while new kinds of jobs also appear.*

How do experts judge a job's automation risk?

Experts look at the small tasks that make up a job. When those tasks are routine, clearly defined, and follow repeated patterns-and there are lots of examples for computers to learn from-AI systems can pick them up more easily.
The shorter and more repetitive a task is, the easier it is to automate.

EXAMPLE JOB	TYPICAL TASK	AI RISK (TODAY)
IT Help Desk Technician	10-minute help IT tickets that follow a script	*Higher Risk* – chatbots can handle most tickets
IT Systems Designer	Plan how the company's tech infrastructure will work over the next year	*Lower Risk* – needs long-term planning and team buy-in

Other factors also influence how likely a job is to be replaced by AI automation:

Jobs are at greater risk when they:

- Can be done entirely online, or
- Have lots of recorded data for AI to learn from.

Jobs are less at risk when they:

- Require in-person skills or dexterity, or
- Rely on human trust or empathy.

Instrument: Expert Source Introduction (Step 3a)

Now let's compare your view with what experts think.

- Dr. Carl Benedikt Frey and Dr. Michael A. Osborne are researchers at the University of Oxford, where they investigate the impact of AI and automation on the labor market. *Their 2017 academic study, "The Future of Employment," is a landmark in this field—cited more than 17,000 times and widely used to gauge automation risk.*
- Here's how Dr. Frey and Dr. Osborne estimated job-automation risk:

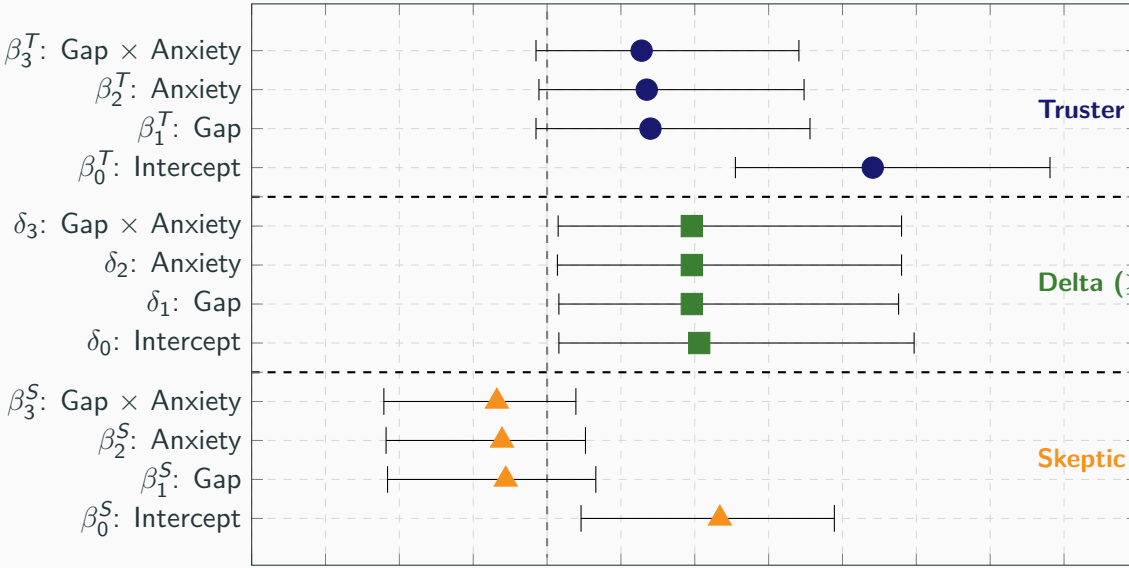
1. They started with the U.S. government's O*NET database, which lists the skills and day-to-day tasks for every job.
2. A panel of AI researchers first tagged 70 sample jobs as either "automatable" or "not yet automatable."
3. A computer model then learned the patterns: jobs heavy on routine, rule-based tasks, fine hand-eye work, or data processing score higher; jobs that rely on creative thinking, social insight, or physical dexterity score lower.

That model produced a score for each job—the higher the number, the sooner experts think machines could take over most of its tasks. *We use that score as the expert estimate for your occupation's relative risk of AI automation.*

Your view vs. the expert estimate

Below, you'll see where you placed your job on the automation-risk scale, compared to the expert estimate for your occupation.

Model Coefficients (Forest Plot)



BFMM Treatment Effects (Full Table)

Parameter	Estimate	SD	95% CI	$P(\beta > 0)$
Panel A: Skeptic Parameters (β_j^S)				
β_0^S: Intercept	0.234	0.105	[+0.047, +0.389]	0.975
β_1^S: Gap	-0.056	0.087	[-0.216, +0.066]	0.261
β_2^S: Anxiety	-0.061	0.085	[-0.218, +0.052]	0.230
β_3^S: Gap $\times$ Anxiety	-0.068	0.082	[-0.221, +0.039]	0.183
Panel B: Trust-Type Differential ($\beta_{\Delta,j} \geq 0$)				
$\beta_{\Delta,0}$: Intercept	0.206	0.153	[+0.016, +0.498]	1.000
$\beta_{\Delta,1}$: Gap	0.196	0.147	[+0.016, +0.476]	1.000
$\beta_{\Delta,2}$: Anxiety	0.196	0.148	[+0.015, +0.480]	1.000
$\beta_{\Delta,3}$: Gap $\times$ Anxiety	0.196	0.148	[+0.016, +0.481]	1.000
Panel C: Truster Parameters ($\beta_j^T = \beta_j^S + \beta_{\Delta,j}$)				
β_0^T: Intercept	0.441	0.131	[+0.255, +0.680]	1.000
β_1^T: Gap	0.140	0.117	[-0.015, +0.356]	0.924
β_2^T: Anxiety	0.135	0.113	[-0.011, +0.348]	0.929
β_3^T: Gap $\times$ Anxiety	0.129	0.112	[-0.014, +0.341]	0.919

Partisan Heterogeneity

Group	ATE (SD)	90% CI	CATEhigh	P(>0.1)
Panel A: Marginal Partisan Effects (ignoring trust class)				
Non-Democrat	0.20	[0.05, 0.34]	—	87%
Democrat	0.35	[0.23, 0.48]	—	100%
Difference (Dem − Non-Dem)	0.15	[-0.04, 0.34]	—	68%
Non-Liberal	0.21	[0.07, 0.34]	—	92%
Liberal	0.35	[0.22, 0.47]	—	100%
Difference (Lib − Non-Lib)	0.14	[0.02, 0.29]	—	67%
Panel B: Cell-Specific Effects (Trust Class × Party)				
Skeptic × Non-Democrat	0.14	[-0.03, 0.30]	-0.21 [-0.56, 0.10]	66%
Skeptic × Democrat	0.29	[0.12, 0.46]	0.31 [-0.05, 0.63]	97%
Truster × Non-Democrat	0.32	[0.12, 0.54]	0.55 [0.12, 1.04]	96%
Truster × Democrat	0.47	[0.17, 0.82]	0.92 [0.28, 1.67]	97%
Panel C: Decomposition (Main Effects and Interaction)				
Trust Main (Truster − Skeptic)	0.18	[-0.07, 0.47]	—	68%
Partisan Main (Dem − Non-Dem)	0.15	[-0.03, 0.34]	—	68%

Mediation Analysis

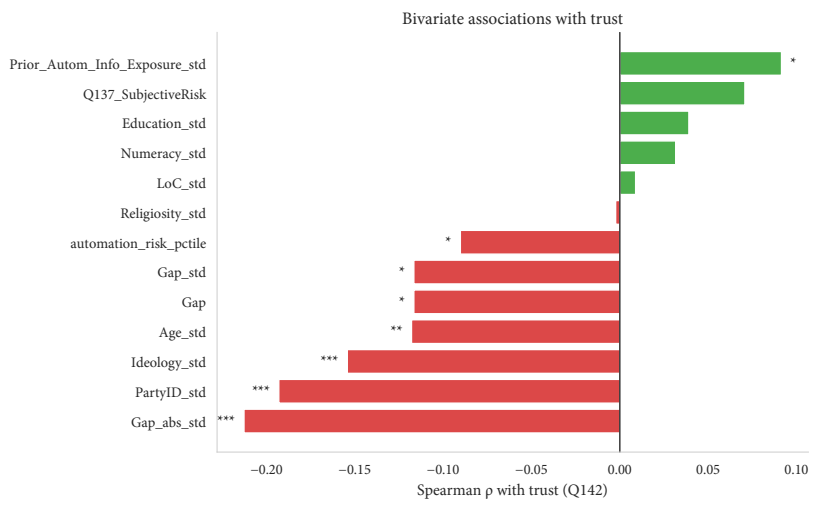
Path	Coefficient	SE	p	90% CI	95% CI
Panel A: Treatment → Mediator (a paths)					
Treatment → AutoAnxiety	0.15*	0.07	.023	[0.04, 0.26]	[0.02, 0.28]
Treatment → EgoThreat	0.24***	0.06	<.001	[0.13, 0.34]	[0.11, 0.36]
Treatment → EgoOpp	−0.04	0.06	.570	[-0.14, 0.07]	[-0.16, 0.09]
Panel B: Mediator → Policy Outcome (b paths, controlling for Treatment)					
AutoAnxiety → Policy	0.04	0.04	.318	[-0.03, 0.11]	[-0.04, 0.12]
EgoThreat → Policy	0.11*	0.05	.011	[0.04, 0.19]	[0.03, 0.20]
EgoOpp → Policy	−0.08*	0.04	.032	[-0.14, -0.02]	[-0.15, -0.01]
Panel C: Direct and Indirect Effects					
Total Effect (c path)	0.23***	0.06	<.001	[0.14, 0.33]	[0.12, 0.35]
Direct Effect (c' path)	0.20***	0.06	<.001	[0.10, 0.29]	[0.08, 0.31]
Indirect Effects (a × b):					
via AutoAnxiety	0.006	—	—	—	—
via EgoThreat	0.027	—	—	—	—

Scale Reliability

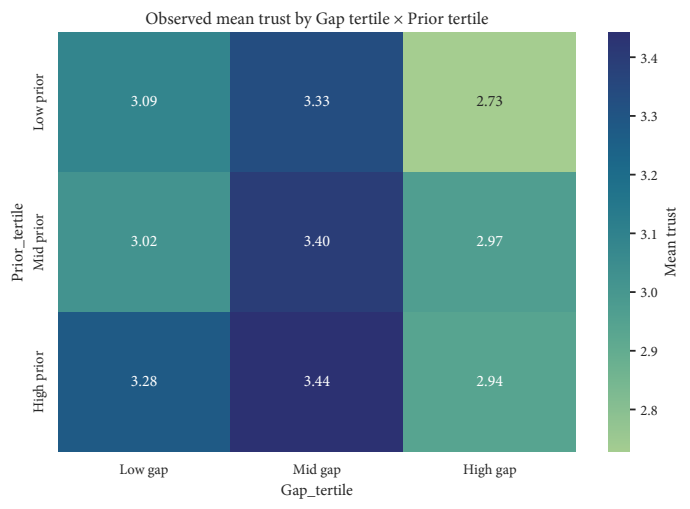
Scale	k	α	ω	CFI	TLI	RMSEA	SRMR	Verdict
Policy Demand (Q96)	5	0.824	0.815	0.983	0.943	0.102	0.027	good fit
Trait Anxiety (Q23)	4	0.914	0.911	0.991	0.944	0.163	0.014	good fit
Automation Anxiety (Q94)	4	0.931	0.921	0.997	0.980	0.110	0.006	good fit
Automation Impact (Q95) — 2-factor	7	—	—	0.988	0.974	0.070	0.027	good fit
Locus of Control (Q81)	4	0.475	0.441	0.896	0.375	0.182	0.042	poor fit
Economic Anxiety (Q24/Q25/Q26)	3	0.705	0.727	1.000	—	—	0.000	good fit
Joint Measurement Model (6 factors, LoC excluded)								
23 items, 6 factors	23	—	—	0.958	0.949	0.054	0.048	good fit

Notes: α = Cronbach's alpha; ω = McDonald's omega (from CFA standardised loadings); CFI = Comparative Fit Index; TLI = Tucker-Lewis Index; RMSEA = Root Mean Square Error of Approximation; SRMR = Standardised Root Mean Square Residual. All CFA models include a priori correlated residuals for item pairs sharing a content sub-theme (see text). Verdict is based on a majority vote among CFI, TLI, and SRMR; χ^2/df and RMSEA are excluded from or down-weighted in the verdict because they are inflated at $N > 300$ (Hu & Bentler 1999; Kline 2016). Thresholds: CFI, TLI ≥ 0.95 (good), ≥ 0.90 (acceptable); SRMR ≤ 0.08 (good), ≤ 0.10 (acceptable). LoC is excluded

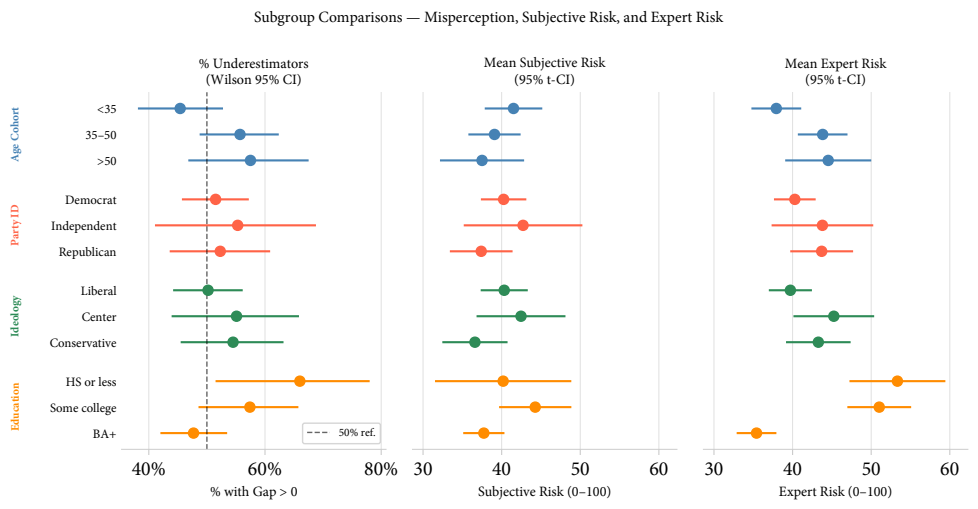
Trust Correlates



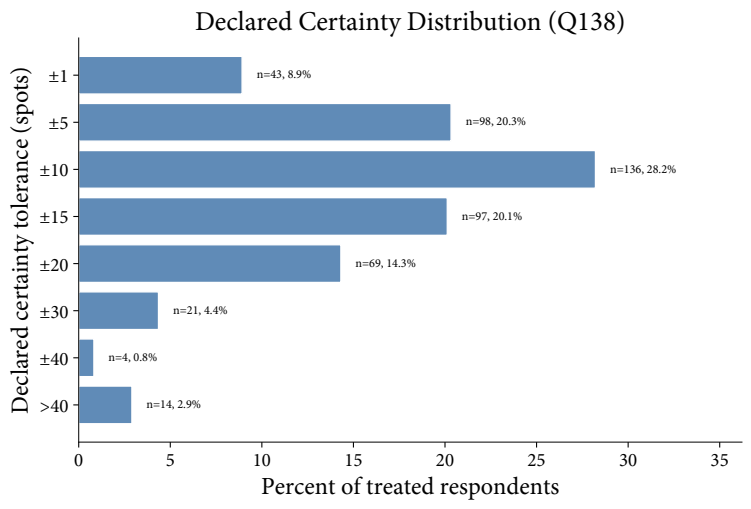
Trust by Gap × Prior Exposure (Raw Means)



Belief Gap Across Demographic Subgroups



Declared Certainty Distribution (Q138)



Prior Distributions

Component	Parameter	Prior	Rationale
Trust Model	π	$\text{Beta}(2, 2)$	Weakly informative, centered at 0.5
	δ	$\mathcal{N}^+(3, 3)$	Positive separation; expects moderate effect
	$\boldsymbol{\kappa}$	$\mathcal{N}(0, 1.5)$	Regularized cutpoints
	$\boldsymbol{\gamma}_{\text{trust}}$	$\mathcal{N}(0, 1)$	Regularized covariate effects
Trait CFA	$\boldsymbol{\nu}_{\text{trait}}$	$\mathcal{N}(3, 1.5)$	Item intercepts (raw 1–7 scale)
	$\lambda_{\text{trait},1}$	$= 1$ (fixed)	Reference indicator
	$\boldsymbol{\lambda}_{\text{trait},2:J}$	$\mathcal{N}^+(0.9, 0.3)$	Positive loadings near unity
	$\eta_{\text{trait},i}$	$\mathcal{N}(0, 1)$	Soft prior (scale stabilization)
Policy CFA	$\boldsymbol{\alpha}_{\text{policy}}$	$\mathcal{N}(4, 2)$	Item intercepts (raw 1–7 scale)
	$\lambda_{\text{policy},1}$	$= 1$ (fixed)	Reference indicator
	$\boldsymbol{\lambda}_{\text{policy},2:J}$	$\mathcal{N}^+(0.8, 0.3)$	Positive loadings, moderate values
Structural	α_{skeptic}	$\mathcal{N}(0, 1)$	Regularized baseline
	α_Δ	$\mathcal{N}(0, 0.5)$, $\alpha_\Delta \geq -0.1$	Tight prior; allows near-zero separation
	$\boldsymbol{\beta}^{(S)}$	$\mathcal{N}(0, 0.5)$	Regularized for realistic effect sizes

Sample Demographics

Variable	Category	*n*	%
Gender			
	Male	425	46.7
	Female	469	51.5
	Non-binary / Other	11	1.2
	Prefer not to say	5	0.5
Race / Ethnicity			
	White	629	69.1
	Black or African American	83	9.1
	Hispanic or Latino	1	0.1
	Asian	72	7.9
	Native Hawaiian / Pacific Islander	4	0.4
	Other	3	0.3
	Prefer not to say	51	5.6
	Multiracial	67	7.4
Education			
	Less than high school	4	0.4

Descriptive Statistics

Variable	*N*	Mean	SD	Min	Median	Max
Panel A: Outcome and Treatment						
Policy Demand (mean, 1–5)	910	5.49	1.15	1.00	5.60	7.00
Policy Demand (std.)	910	-0.00	1.00	-3.91	0.09	1.31
Treatment Assignment	910	0.53	0.50	0.00	1.00	1.00
Panel B: Psychological Scales						
Trait Anxiety (mean, 1–7)	910	3.65	1.60	1.00	3.75	7.00
Automation Anxiety (mean, 1–7)	910	3.29	1.67	1.00	3.00	7.00
Automation Impact F1 (mean, 1–5)	910	3.11	1.07	1.00	3.25	5.00
Automation Impact F2 (mean, 1–5)	910	3.19	0.94	1.00	3.33	5.00
Economic Anxiety (index, 1–5)	910	2.79	0.93	1.00	2.67	5.00
Locus of Control (index, 1–5)	910	3.44	0.70	1.00	3.50	5.00
Numeracy (correct / 3)	910	2.48	0.74	0.00	3.00	3.00
Panel C: Treatment-Specific Variables						
Trust in Expert Source (1–5)	482	3.13	0.95	1.00	3.00	5.00
Subjective Risk (						
Prior Uncertainty (σ)	468	11.8	7.4	1.0	10.0	40.0

Full Balance — Continuous Covariates

	Treated ($n = 482$)		Control ($n = 428$)			
Covariate	Mean	SD	Mean	SD	Δ	SMD
Age (years)	40.16	10.79	38.79	10.43	1.37	0.13
Ideology (1=very liberal, 7=very conservative)	3.19	1.77	3.36	1.85	-0.17	-0.09
Education (1=less than HS, 5=graduate degree)	3.73	0.91	3.64	0.90	0.09	0.10
Numeracy score (0–3)	2.50	0.71	2.46	0.77	0.04	0.06
Party ID (1=strong Dem, 7=strong Rep)	3.10	2.09	3.19	2.15	-0.09	-0.04
Locus of control — internal index	3.43	0.71	3.44	0.70	-0.01	-0.01
Trait anxiety (mean of 4 items)	3.76	1.61	3.53	1.59	0.23*	0.14
Economic anxiety index (std.)	0.03	1.03	-0.05	0.97	0.08	0.08
Expert automation risk score (0–100)	42.29	30.69	40.81	30.72	1.49	0.05
Female (0/1)	0.52	0.50	0.51	0.50	0.00	0.01
White (0/1)	0.70	0.46	0.68	0.47	0.02	0.05
Risk tolerance index	-0.05	0.98	-0.01	1.01	-0.04	-0.04
Patience index	0.01	0.99	0.01	1.03	-0.01	-0.01

Note: Δ = treated minus control mean. SMD = standardized mean difference (Cohen's *d*, pooled

Full Balance — Categorical Covariates

	Treated		Control	
Category	*n*	%	*n*	%
Gender [$\chi^2(3) = 0.65,\ V = 0.027$]				
Male	223	46.3	202	47.2
Female	249	51.7	220	51.4
Other	7	1.5	4	0.9
Prefer not to say	3	0.6	2	0.5
Religious attendance [$\chi^2(5) = 1.48,\ V = 0.040$]				
Every week	39	8.1	37	8.6
Almost every week	18	3.7	19	4.4
1–2/month	30	6.2	26	6.1
Few/year	83	17.2	79	18.5
Never (religious)	71	14.7	53	12.4
Never (not religious)	241	50.0	214	50.0
Prior info about automation risk [$\chi^2(1) = 0.00,\ V = 0.002$]				
No	287	59.5	254	59.3

Scale Inter-Correlations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) Policy Demand	—						
(2) Trait Anxiety	0.16***	—					
(3) Automation Anxiety	0.17***	0.21***	—				
(4) Auto Impact F1	0.27***	0.23***	0.72***	—			
(5) Auto Impact F2	−0.23***	−0.16***	−0.12***	−0.35***	—		
(6) Locus of Control	−0.13***	−0.34***	−0.22***	−0.26***	0.19***	—	
(7) Economic Anxiety	0.25***	0.47***	0.38***	0.42***	−0.26***	−0.40***	—

Notes: Spearman rank correlations, listwise complete ($N = 910$). Scale scores: Policy Demand and Trait Anxiety are item means; Automation Anxiety is the mean of four items (item 4 reverse-coded); Automation Impact F1 and F2 are subscale means from the two-factor Q95 model; Locus of Control is the 4-item internal index (items 3–4 reverse-coded); Economic Anxiety is the mean of Q24, Q25, and reverse-coded Q26. Significance: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

Trust Mixture Covariates

Covariate	Mean	95% CI (low)	95% CI (high)
Gap (linear)	-0.144	-0.317	0.026
Gap^2 (quadratic)	-0.391	-0.553	-0.243
Prior exposure (linear)	0.043	-0.173	0.272
Prior exposure^2 (quadratic)	0.294	0.026	0.566
Class separation δ	2.085	0.304	3.947

Note: Coefficients are gamma parameters from the trust (Q142) measurement model in the BFMM, with Gap and prior exposure standardized as in the main text.

What Predicts the Belief Gap?

Predictor	$\hat{\beta}$	SE (HC3)	p
Ideology (std.)	0.415	1.747	0.812
Education (std.)	-6.383***	1.539	0.000
Age (std.)	3.823*	1.717	0.026
Female	1.665	3.408	0.625
White	0.933	3.433	0.786
Numeracy (std.)	2.482	2.214	0.263
Economic anxiety (std.)	-2.528	1.936	0.192
Trait anxiety (std.)	-1.030	2.009	0.608
Locus of control (std.)	3.274	1.883	0.083
Prior automation info (std.)	-7.367***	1.598	0.000
Constant	0.666	3.191	0.835

Note: OLS regression, $n = 482$. $R^2 = 0.129$, adjusted $R^2 = 0.110$. $F(10, 471) = 6.98$, $p = 0.000$. SE = heteroskedasticity-robust (HC3). Dependent variable: belief gap (expert score − respondent subjective score). All continuous predictors standardized. Stars: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

Full Manipulation Checks

Panel A. Q141 Response Distribution and Perception–Gap Alignment

Response category	n	%
Significantly lower risk	96	20.2
Slightly lower risk	72	15.2
Similar risk	70	14.7
Slightly higher risk	97	20.4
Significantly higher risk	140	29.5
Spearman r(Q141, Gap) $= 0.737$, $p < 0.001$		
Direction agreement (excluding "similar"): 90.1%		

Treated sample only ($N = 475$).

Full Manipulation Checks (Panels B–C)

Panel B. Mean Q141 by Objective Gap Tercile

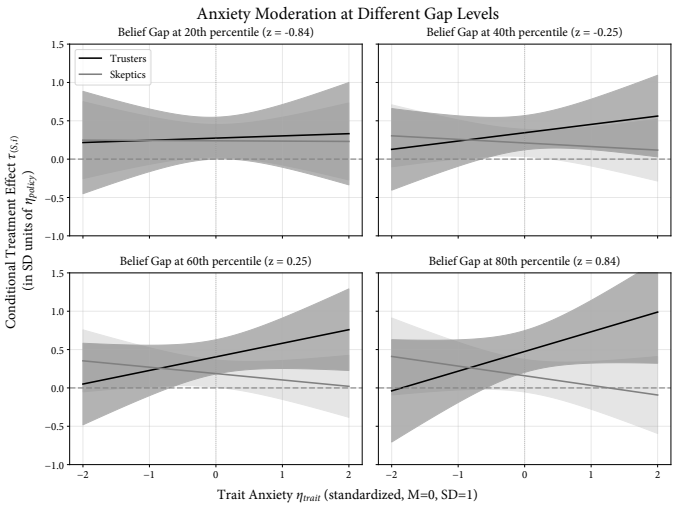
Gap tercile	Mean	SD	*n*
Expert < Subjective	1.92	1.29	158
Mid Gap	3.18	1.00	158
Expert > Subjective	4.61	0.76	159

Panel C. Objective vs. Perceived Direction of the Gap

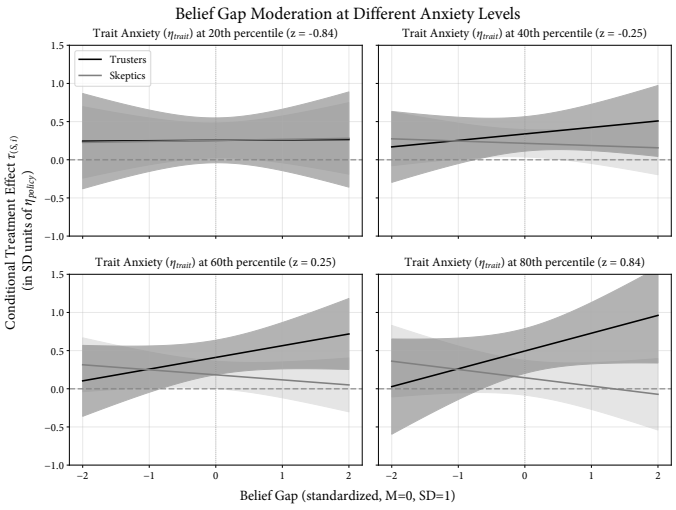
	Perceived Higher	Perceived Lower	Similar	Total
Objective Higher	169	5	11	185
Objective Lower	28	131	14	173
Objective Similar	40	32	45	117
Total	237	168	70	475

Q141 reverse-coded so higher values indicate "experts say higher risk."

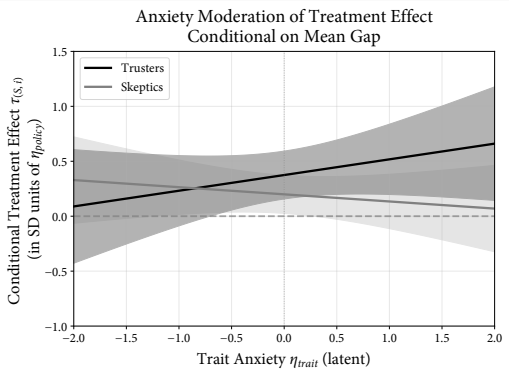
Anxiety Partial Effects (Continuous Ribbons)

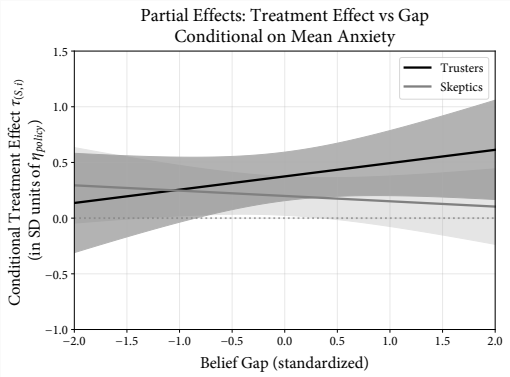


Gap Partial Effects (Continuous Ribbons)

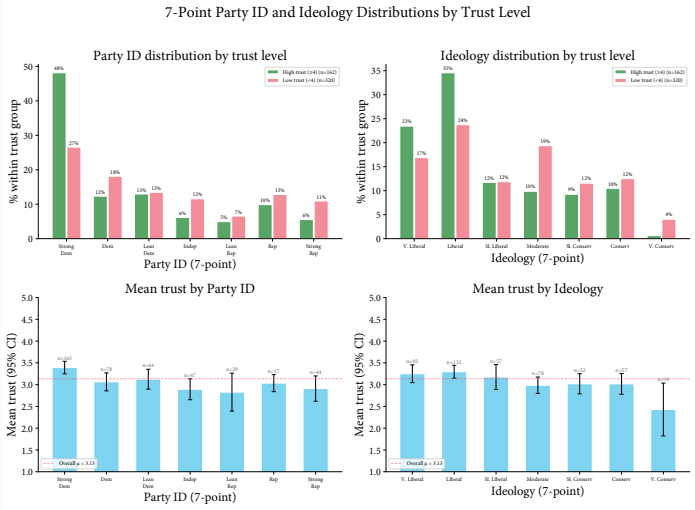


Single-Moderator Partial Effects





Trust by 7-Point Party ID and Ideology



Automation Anxiety Item Effects

Item	Control	Treatment	β	SE	p	90% CI
Q94_1: Job automated in 5 years	2.84	3.00	0.08	0.07	.205	[-0.02, 0.19]
Q94_2: Job automated in 10 years	3.26	3.59	0.16*	0.07	.015	[0.05, 0.27]
Q94_3: Concerned about losing job	3.03	3.41	0.17**	0.06	.007	[0.07, 0.28]
Q94_4: Feel secure (reversed)	3.44	3.69	0.12†	0.07	.077	[0.01, 0.22]

Notes: Structural Equation Model with correlated residuals within scale and full covariate vector (ideology, party ID, age, gender, race, education, religiosity, locus of control, numeracy, trait anxiety). Items measured on 1–7 scale (Strongly Disagree to Strongly Agree). Control and Treatment columns show group means. β is the standardized treatment effect. $N = 910$.

Q94_4 ("I feel secure that my occupation is safe from automation") is reverse-coded; higher values indicate *less* security.

† $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Perceived AI Impact Item Effects

Item	Control	Treatment	β	SE	p	90% CI
Threat Items (items 1, 3, 5, 7)						
Q95_1: Reduced job opportunities	3.15	3.41	0.21**	0.06	.001	[0.10, 0.31]
Q95_3: Lower salary	2.95	3.18	0.20**	0.06	.002	[0.09, 0.30]
Q95_5: Decreased job security	2.96	3.27	0.23***	0.06	.000	[0.13, 0.34]
Q95_7: Worsened conditions	2.82	3.08	0.21***	0.06	.001	[0.11, 0.32]
Opportunity Items (items 2, 4, 6)						
Q95_2: New job opportunities	2.88	2.88	0.03	0.06	.651	[-0.08, 0.14]
Q95_4: Enhanced satisfaction	3.16	3.07	-0.05	0.06	.428	[-0.16, 0.05]
Q95_6: Enhanced productivity	3.61	3.54	-0.07	0.07	.271	[-0.18, 0.04]

Notes: Structural Equation Model with correlated residuals within subscale and full covariate vector (ideology, party ID, age, gender, race, education, religiosity, locus of control, numeracy, trait anxiety). Items measured on 1–5 scale. Control and Treatment columns show group means. β is the standardized treatment effect. $N = 910$.

† $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Full Mediation Path Decomposition

Path	Coefficient	SE	p	90% CI	95% CI
Panel A: Treatment → Mediator (a paths)					
Treatment → AutoAnxiety	0.15*	0.07	.023	[0.04, 0.26]	[0.02, 0.28]
Treatment → EgoThreat	0.24***	0.06	<.001	[0.13, 0.34]	[0.11, 0.36]
Treatment → EgoOpp	−0.04	0.06	.570	[-0.14, 0.07]	[-0.16, 0.09]
Panel B: Mediator → Policy Outcome (b paths, controlling for Treatment)					
AutoAnxiety → Policy	0.04	0.04	.318	[-0.03, 0.11]	[-0.04, 0.12]
EgoThreat → Policy	0.11*	0.05	.011	[0.04, 0.19]	[0.03, 0.20]
EgoOpp → Policy	−0.08*	0.04	.032	[-0.14, -0.02]	[-0.15, -0.01]
Panel C: Direct and Indirect Effects					
Total Effect (c path)	0.23***	0.06	<.001	[0.14, 0.33]	[0.12, 0.35]
Direct Effect (c' path)	0.20***	0.06	<.001	[0.10, 0.29]	[0.08, 0.31]
Indirect Effects (a × b):					
via AutoAnxiety	0.006	—	—	—	—
via EgoThreat	0.027	—	—	—	—

Treatment Effects on Mechanism Scales

Outcome	β	SE	p	90% CI	95% CI
Panel A: Mechanism Scales					
Automation Anxiety (Q94)	0.167*	0.066	.012	[0.06, 0.28]	[0.04, 0.30]
Egotropic Threat (Q95)	0.248***	0.066	$<$.001	[0.14, 0.36]	[0.12, 0.38]
Egotropic Opportunity (Q95)	$-$0.054	0.066	.418	[-0.16, 0.06]	[-0.18, 0.08]
Panel B: Primary Outcome (for comparison)					
Policy Demand (Q96)	0.275***	0.066	$<$.001	[0.17, 0.38]	[0.14, 0.40]

Notes: OLS regression with HC3 robust standard errors. All outcomes standardized (mean $= 0$, SD $=$ 1). β represents the treatment effect in standard deviation units.

Automation Anxiety: 4-item scale measuring job automation concerns (Q94_1–4, with Q94_4 reverse-coded). Egotropic Threat: 4-item scale measuring perceived negative AI career impacts (Q95_1, Q95_3, Q95_5, Q95_7). Egotropic Opportunity: 3-item scale measuring perceived positive AI career impacts (Q95_2, Q95_4, Q95_6).

* $p < .05$; ** $p < .01$; *** $p < .001$.

Partisan Effects vs. Pre-Registered Criteria

Effect	Est.	HDI $\neq 0$?	P(>0.05)	Decision
Pass 1: Marginal Effects				
Democrat − Non-Democrat	+0.15 SD	Marginal	82%	Suggestive
Liberal − Non-Liberal	+0.14 SD	Yes	85%	Suggestive
Pass 2–3: Interaction Model				
Trust Main Effect	+0.18 SD	No	78%	Suggestive
Partisan Main Effect	+0.15 SD	No	82%	Suggestive
Trust × Partisan	+0.01 SD	No	41%	**Null**
Within-Class Contrasts				
Within Skeptics	+0.15 SD	No	—	Suggestive
Within Trusters	+0.16 SD	No	—	Suggestive

Note: "Suggestive" = consistent positive direction but fails preregistered criteria (90% HDI excludes zero *and* P(>0.05)> 0.95). "Null" = no directional evidence. P(>0.05) computed from posterior normal approximation.

Partisan × Trust Heterogeneity

Group	ATE (SD)	90% CI	CATEhigh	P(>0.1)
Panel A: Marginal Partisan Effects (ignoring trust class)				
Non-Democrat	0.20	[0.05, 0.34]	—	87%
Democrat	0.35	[0.23, 0.48]	—	100%
Difference (Dem − Non-Dem)	0.15	[-0.04, 0.34]	—	68%
Non-Liberal	0.21	[0.07, 0.34]	—	92%
Liberal	0.35	[0.22, 0.47]	—	100%
Difference (Lib − Non-Lib)	0.14	[0.02, 0.29]	—	67%
Panel B: Cell-Specific Effects (Trust Class × Party)				
Skeptic × Non-Democrat	0.14	[-0.03, 0.30]	-0.21 [-0.56, 0.10]	66%
Skeptic × Democrat	0.29	[0.12, 0.46]	0.31 [-0.05, 0.63]	97%
Truster × Non-Democrat	0.32	[0.12, 0.54]	0.55 [0.12, 1.04]	96%
Truster × Democrat	0.47	[0.17, 0.82]	0.92 [0.28, 1.67]	97%
Panel C: Decomposition (Main Effects and Interaction)				
Trust Main (Truster − Skeptic)	0.18	[-0.07, 0.47]	—	68%
Partisan Main (Dem − Non-Dem)	0.15	[-0.03, 0.34]	—	68%